

Shanilka Haturusinghe

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EDUCATION

University of Kelaniya, Sri Lanka | *BSc (Hons) in Computer Science, First Class*
2020 – 2024

- GPA: 4.0/4.0 (*Ranked 1st in the Faculty of Computing and Technology*)
- Specialization: Artificial Intelligence
- Awarded two gold medals in 2025 for Academic Excellence and Outstanding Performance by the Alumni Association of the University of Kelaniya and the Faculty of Computing and Technology, respectively.

Royal College, Colombo | *G.C.E. Advanced Level, Mathematics Stream (English Medium)*
2018

- Results: A in Combined Mathematics, B in Physics, B in Chemistry

PUBLICATIONS

- **Shanilka Haturusinghe**, Tharindu Cyril Weerasooriya, Christopher M. Homan, Marcos Zampieri, and Sidath Ravindra Liyanage. 2025. *Subasa – Adapting Language Models for Low-resourced Offensive Language Detection in Sinhala*. In *Proceedings of NAACL 2025: Student Research Workshop*, pages 260–270, Albuquerque, USA. ACL. DOI: 10.18653/v1/2025.naacl-srw.26
- U. P. Melder, **S. Haturusinghe**, and S. P. K. Arachchi. 2026. *Customer Segmentation and Churn Prediction in the Maritime Shipping Industry Using Machine Learning Techniques: A Sri Lankan Case Study*. In *2026 6th International Conference on Advanced Research in Computing (ICARC)*, Belihuloya, Sri Lanka, pages 1–6. DOI: 10.1109/ICARC68737.2026.11453542

TEACHING EXPERIENCE

University of Kelaniya

Lecturer (Temporary)

Colombo, Sri Lanka

August 2025 – Present

- Lecturer-in-Charge for CSCI 31032 - **Theory of Programming Languages**: delivered all lectures, office hours, assignments, grading, and examinations.
- Contributing lecturer for CSCI 31022 - **Machine Learning and Pattern Recognition** and CSCI 11532 - **Structured Programming I**.

Teaching Assistant & Demonstrator

February 2025 – July 2025

- Served as TA for DSCI 32012 - **Advanced Database Applications** (70 students), CTEC 22043 - **Object Oriented Programming** (50 students), AINT 32012 - **Digital Image Processing and Computer Vision** (10 students), and CSCI 22082 - **Object-Oriented Programming II** (50 students).
- Conducted office hours, led lab sessions, and developed and graded assignments across all modules.

INDUSTRY EXPERIENCE

Embla Software Innovation (Pvt) Ltd

Colombo, Sri Lanka

Software Engineer Intern

October 2023 – April 2024

- Developed and implemented the "Scrapbook" feature for the Embla FT genealogy application with customizable page templates, editable text/photo elements, and comprehensive image and content management capabilities.
- Built full-stack components using C#/.NET Core backend and Angular 9/TypeScript frontend, enhancing existing architecture to support new functionality and user interactions.
- Collaborated with product teams using Agile methodologies, Jira, and Bitbucket; utilized Visual Studio, FastReport Designer, and Figma for development and design.

Xemplar Pte. Ltd

Remote

Intern – Software Developer

September 2022 – May 2023

- Designed and implemented an image processing pipeline for Sri Lanka License Plate Detection using Python, TensorFlow, and YOLO models; applied OpenCV preprocessing techniques and integrated Tesseract OCR for alphanumeric text extraction.
- Designed and implemented a real-time chat feature for the IFAWPCA mobile app using React Native, integrating Firebase Realtime Database and Firebase Authentication for instant message synchronization and user management.

ACHIEVEMENTS & AWARDS

Gold Medal for Academic Excellence – Alumni Association of the University of Kelaniya 2025

Gold Medal for Outstanding Performance – Faculty of Computing and Technology, University of Kelaniya 2025

Google Cloud Research Credits (Award GCP19980904) – \$1,000 in research credits 2024

Huawei Seeds for the Future – Selected as one of four top university students from Sri Lanka to visit China; received mentorship from industry experts on 5G, AI, Cloud Computing, and Digital Power 2023

PROJECTS

PersonaCraft.AI – Personalized AI Writing Assistant | github.com/haturusinghe/persona-craft-ai

- Developed an end-to-end system for creating personalized AI writing assistants through automated data collection, processing, and model fine-tuning with multi-platform web scraping capabilities.
- Built automated data extraction pipelines using Selenium for LinkedIn, Medium, GitHub, and custom web sources, with advanced text processing using chunking and embedding generation via Sentence Transformers.
- Engineered training pipeline supporting both Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO) using TRL and Unsloth; implemented MLOps architecture with ZenML, MongoDB, Qdrant, and AWS SageMaker.
- Key Technologies: Python, ZenML, LangChain, GPT-4, Llama 3.1, TRL, Unsloth, MongoDB, Qdrant, AWS SageMaker, Docker

Real-Time Fashion Recommendation System | github.com/haturusinghe/realtime-fashion-recommender

- Developed an MLOps-driven fashion recommendation system using H&M's Kaggle dataset, implementing a 4-stage decoupled pipeline architecture for feature processing, training, and

inference.

- Implemented a two-tower neural network for customer and item embedding generation; used CatBoost for ranking model training and Hopsworks Feature Store for embedding storage and retrieval.
- Integrated Hopsworks Model Registry and Model Serving for deployment, with an interactive Streamlit frontend for real-time customer recommendations and feedback collection.
- Key Technologies: Python, TensorFlow, Hopsworks, Sentence-Transformers, CatBoost, Polars, Streamlit

Deep Learning-based Real-Time Facial Age and Gender Detection System

- Designed and trained CNN models (separable convolutions, batch normalization, LeakyReLU, dropout) on the UTKFace dataset with data augmentation, achieving robust age regression and binary gender classification.
- Built an end-to-end inference pipeline with TensorFlow Lite supporting CPU, GPU, and NNAPI acceleration for on-device real-time inference on Android.
- Key Technologies: Python, TensorFlow, TF-Lite, Flask, Kotlin, Android Studio, OpenCV

SKILLS & TECHNOLOGIES

Programming Python, C#, TypeScript, Kotlin, L^AT_EX

ML & Deep Learning PyTorch, TensorFlow, Keras, Scikit-learn

LLMs & NLP Hugging Face Transformers, LangChain, LangGraph, Llama, Mistral, LoRA/QLoRA, SFT, DPO, RAG

MLOps & Experiment Tracking ZenML, AWS SageMaker, Weights & Biases, Docker

Data & Databases MongoDB, Qdrant, Pandas, Polars, PostgreSQL

Web & APIs FastAPI, Flask, Streamlit

Dev Tools Git, GitHub, Bitbucket, Jira

REFERENCES

Prof. S. R. Liyanage Professor, Department of Software Engineering, Faculty of Computing and Technology, University of Kelaniya
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